

Feng-Kai (Leonel) Huang

PH.D. STUDENT · NATIONAL TAIWAN UNIVERSITY

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Summary

Feng-Kai (Leonel) Huang is a Ph.D. student in Computer Science and Information Engineering at National Taiwan University (NTU), where he is a member of the Communications and Multimedia Laboratory (**CMLab**) in the Artificial Intelligence and MultiMedia (**AIMM**) Research Group. His research is supervised by Prof. **Wen-Huang Cheng** and Prof. **Hong-Han Shuai**. His consistent academic excellence is highlighted by his induction into the Phi Tau Phi Scholastic Honor Society twice during his undergraduate and master's studies.

His research interests are centered on **Artificial Intelligence, Machine Learning, Computer Vision**, and **Multimedia Processing**. He is passionate about developing innovative algorithms to solve real-world challenges in computer vision, with publications in international conferences such as ACM MM, ICCV and WACV. He is always enthusiastic about exploring new ideas and open to potential research collaborations.

Education

NTU (National Taiwan University)

Taipei, Taiwan

PH.D. STUDENT IN COMPUTER SCIENCE AND INFORMATION ENGINEERING

Aug. 2024 - Present

- GPA: 4.26/4.3 (Fall Semester 2024-2025).
- Advisor: Prof. **Wen-Huang Cheng** and Prof. **Hong-Han Shuai**.
- Laboratory: Communications and Multimedia Laboratory (**CMLab**) - Artificial Intelligence and Multimedia (**AIMM**) Research Group.

NYCU (National Yang Ming Chiao Tung University)

Hsinchu, Taiwan

MASTER OF THE GRADUATE DEGREE PROGRAM OF ARTIFICIAL INTELLIGENCE

Sep. 2022 - Jul. 2024

- GPA: 4.25/4.3.
- Advisor: Prof. **Hong-Han Shuai**.
- Laboratory: Big data Analytics and Social Intelligent Computing (**BASIC**) Laboratory.

CYCU (Chung Yuan Christian University)

Taoyuan, Taiwan

BACHELOR OF ELECTRICAL ENGINEERING

Sep. 2018 - Jun. 2022

- GPA: 3.98/4.
- Rank: 1/104 (ranked in department).

Publications

CONFERENCE

- **F.-K. Huang**, B.-L. Huang, L.-W. Tsao, J.-C. Wu, H.-H. Shuai, and W.-H. Cheng. "Flowing Crowd to Count Flows: A Self-Supervised Framework for Video Individual Counting". In: *ACM International Conference on Multimedia (MM)*. 2025. (Regular Track Paper) [\[Website\]](#) [\[Code\]](#)
- **F.-K. Huang**, H.-W. Xu, C.-C. Lee, H.-Y. Tu, H.-H. Shuai, and W.-H. Cheng. "OinkTrack: An Ultra-Long-Term Dataset for Multi-Object Tracking and Re-Identification of Group-Housed Pigs". In: *ACM International Conference on Multimedia (MM)*. 2025. (Dataset Track Paper) [\[Website\]](#) [\[Code\]](#)
- B.-L. Huang, Z.-X. Ni, **F.-K. Huang**, H.-H. Shuai, and W.-H. Cheng. "When Anchors Meet Cold Diffusion: A Multi-Stage Approach to Lane Detection". In: *International Conference on Computer Vision (ICCV)*. 2025. (Accepting rate = 24% (2698/11239)) [\[Code\]](#)
- C.-Y. Chiu, **F.-K. Huang**, T.-F. Hsiao, H.-H. Shuai, and W.-H. Cheng. "IterDiff: Training-Free Iterative Face Editing via Efficient CLIP-guided Memory Bank". In: *IEEE International Conference on Image Processing (ICIP)*. 2025. [\[Paper\]](#) [\[Code\]](#)
- C.-L. Wan, **F.-K. Huang**, and H.-H. Shuai. "Density-Based Flow Mask Integration via Deformable Convolution for Video People Flux Estimation". In: *IEEE Winter Conference on Application of Computer Vision (WACV)*. 2024. [\[Paper\]](#) [\[Code\]](#)
- Y.-P. Liao, **F.-K. Huang**, Y.-J. Xia, and H. Cheng. "Smart Speaker Based on Detection of Millimeter Wave". In: *IEEE International Conference on Consumer Electronics-Taiwan (ICCE-TW)*. 2022. [\[Paper\]](#)

UNDER REVIEW/IN PREPARATION

- **F.-K. Huang**, K.-J. Weng, J.-C. Wu, H.-H. Shuai, and W.-H. Cheng. "Density-Assisted Adaptive Alignment Network for Video Individual Counting". In: *Submitted to ACM Transactions on Intelligent Systems and Technology (TIST)* (2025)

Honors & Awards

ACADEMIC HONORS

2024	Graduate Research Fellowship , National Science and Technology Council [Link]	NTU
2024	Honorary Member , The Phi Tau Phi Scholastic Honor Society of the Republic of China [Link]	NYCU
2022	Certificate of Excellence , Spring Semester 2022-2023 [Link]	NYCU
2022	Honorary Member , The Phi Tau Phi Scholastic Honor Society of the Republic of China [Link]	CYCU
2018	Certificate of Excellence , Fall & Spring Semesters 2018-2019, Fall & Spring Semesters 2019-2020, Fall & Spring Semesters 2020-2021 [Link]	CYCU
2018	Subjects Merit Awards , Calculus I, Circuit Theory I & II, Engineering Mathematics II, Electronics II, The Principles of Communications, Business Data Communication, Electromagnetism [Link]	CYCU

INTERNATIONAL AWARDS

2021	Gold Prize , 2021 Seoul International Invention Fair Korea Invention Promotion Association [Link]	Online
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DOMESTIC AWARDS

2022	Honorable Mention Award , 2022 AI CUP Competition on UAV Smart Counting Ministry of Education [Link]	Taipei, Taiwan
2021	Honorable Award , 2021 Student Engineering Papers Competition Chinese Institute of Engineers [Link]	Online
2021	1st Place (TQC+) , International ICT Innovative Services Awards 2021 Ministry of Economic Affairs, Ministry of Education, Chinese Society for Information Management [Link]	Online
2021	3rd Place (IP) , International ICT Innovative Services Awards 2021 Ministry of Economic Affairs, Ministry of Education, Chinese Society for Information Management [Link]	Online
2020	3rd Place (TQC+) , International ICT Innovative Services Awards 2020 Ministry of Economic Affairs, Ministry of Education, Chinese Society for Information Management [Link]	Taipei, Taiwan

Work Experience

National Taiwan University

TEACHING ASSISTANT – ARTIFICIAL INTELLIGENCE (SPRING SEMESTER 2024-2025)	Taipei, Taiwan Mar. 2025 - Jun. 2025
- Co-designed three hands-on assignments covering MLLM, RAG, and Multi-Agent Systems, plus an open-ended final project. - Refined each assignment through self-testing to ensure accessibility and challenge for students of all levels. - Managed all assessment activities.	

Patents

- Y.-P. Liao, **F.-K. Huang**, Y.-J. Xia, and H. Cheng. “Rotate-type smart speaker and rotate-type home appliance controller”. TWI817194B. C. Y. C. University. Oct. 1, 2023 [\[Link\]](#)
- F.-K. Huang**, Y.-J. Xia, and H. Cheng. “System for detecing fecal and toilet lid with fecal detection thereof”. TWI788834B. C. Y. C. University. Jan. 1, 2023 [\[Link\]](#)

Skills

Programming	Python, C/C++, LaTeX, Verilog
Research Interest	Artificial Intelligence, Machine Learning, Computer Vision, Multimedia Processing, Generative Models
Developer Tools	Jetson Nano, Raspberry Pi, Arduino
Languages	Mandarin (native), English (TOEIC 800/990)